

Tennessee Tip to Tip — Statewide Build-Out Strategy

Civvix Inc.

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What it takes to go from four proven counties to a statewide service any jurisdiction can sign up for, delivered quarterly, with business-license and TPP support bundled on top.

Where this stands today

Four counties run through one config-driven pipeline in two days: Wilson (88k rooftops), Sumner (107k), Rutherford (180k), Davidson (463k). Every one produced a placed rooftop surface, seams measured in feet, a ranked exception queue, a per-county workbench, and a bulletin — from free data, with one calibration against a real coded roster. The statewide layers are already in the warehouse: 473 DOR situs polygons (every code in Tennessee), 345 certified city boundaries, 95 county polygons. Secretary of State entity files for ~75 counties are already in hand. **The method is proven; what remains is scale, automation and operations.**

What “tip to tip” consists of

Four assets, each built once, then maintained:

1. **The statewide surface** — every rooftop in Tennessee placed in its DOR polygon, measured to its nearest seam, cross-checked against the Comptroller and 911 layers, and scored. Roughly 3.3 million address points across 95 counties. Refreshed quarterly.
2. **The business universe** — every active for-profit entity (SOS) plus every 911 business label, city roster and permit layer that exists, geocoded to a rooftop. Refreshed quarterly.
3. **The jurisdiction views** — 95 county workbenches and 345 city workbenches cut from the same surface (a city’s view is the same data filtered to its situs code, with both directions shown).
4. **The operating loop** — roster intake, adjudication, correction packets, bulletins, calibration.

The build, by phase

Phase 1 — statewide surface (weeks 1-4)

Task	How	Effort
Pull NG911 address points + roads for 91 remaining counties	<code>fetch_county.py</code> - - live on a machine with internet (GitHub Actions or a \$20/mo VM); ~3M points, ~1.5 GB	1 engineer-week, mostly wall-clock
Generate 91 county configs	Script: situs codes and names	1 engineer-week

Task	How	Effort
	come straight from the DOR layer; USPS-only places derived automatically (below); city layers discovered by ArcGIS Online search per county	
Automate the postal-county map	For every Post_City + ZIP pair across all 3M rooftops, the majority county is the postal home county. Replaces every hand-typed list; also produces the multi-county-city exceptions (Goodlettsville, Portland, White House) automatically	2 days
Consolidated-government rules	Only Davidson (Metro) and Moore (Lynchburg-Moore) need situs_merge; Hartsville-Trousdale is the third	1 day
Run all 95; publish 95 free boundary reports	Batch; ~6 hours compute	automatic

Deliverable: the free tier live for every county — the lead-generation engine.

Phase 2 — business universe and municipal views (weeks 4-8)

Task	How
SOS entity files for all 95 counties as the primary business layer	Already have ~75; obtain the rest through Civvix's existing SOS channel
Discover per-county 911 business labels, city rosters, permit layers	ArcGIS Online search script, reviewed by hand; Hendersonville-style situs rosters are gold when found
345 city workbenches	Same pipeline, - - jurisdiction city:8305; shows the city's coded-vs-measured both ways
Parcel join where free (86 Comptroller counties + Davidson, Montgomery, Rutherford)	Adds parcel ID to the evidence packet and +5 to the score

Phase 3 — operations (weeks 6-12)

Task	How
Scheduled refresh	GitHub Actions cron (weekly signal poll; quarterly bulk); snapshots to S3; catalog in git
Roster intake	The workbench's browser-side join for small jurisdictions; a county-side script for big ones. Never on Civvix infrastructure
Adjudication capacity	Calibration says ~0.3% of placed licensed

Task	How
	businesses are miscoded. Statewide (~350k active entities) that's ~1,000–1,500 findings a year at full penetration. One analyst handles that; two at scale
Correction packets + letters	Generated from the workbench; county signs and sends
Calibration reports	After each jurisdiction's first roster; published to the client
Financial Control relationship	One person owns it; the same three questions (contractor designation, lookback authority, PO Box rule) answered once for everyone

Phase 4 — the annual license and TPP check (parallel, from week 8)

The situs surface is also the denominator Civvix has been missing. Every business is now a rooftop with a jurisdiction. That answers, per business:

- **Business license** — which county *and* which city should have issued one (T.C.A. § 67-4-708), and whether the clerk's roster shows it. The SOS entity file against the clerk's roster, placed on the map, is the discovery product Civvix already sells — now with jurisdiction certainty.
- **TPP** — which assessor's roll the personalty belongs on (§ 67-5-502 situs; home-base rule), and whether a schedule was filed. Same placed universe against the assessor's TPP roll.

One annual run per jurisdiction: placed universe × license roster × TPP roll → three exception lists from one geography. The situs subscription is the quarterly relationship; the annual check is the recovery event. They share every byte of infrastructure.

What it costs

	Year 1	Ongoing
Engineering (Marty + one)	the phases above, ~3 engineer-months	0.5 FTE maintenance
Analysts	1 from month 3	2 at 40+ jurisdictions
Infrastructure	S3 + a small VM/Actions: under \$500/mo at full scale	same
Data	\$0 for the surface; ~\$2–3K one-time reserve for the three or four parcel gaps	\$0 recurring
Insurance	~\$3–5K/yr (see insurance memo)	same
Legal	Tennessee counsel on the § 67-1-1704(e) reuse question and the engagement letter — once	—

Sequencing that compounds

Start from the Wilson seam outward. Every county line has two sides, and a finding on one side is a finding on the other: the Lebanon Road / Old Hickory corridor is simultaneously a Wilson and a Davidson matter; Brentwood's ZIP spilling into south Davidson (7,481 rooftops) is a Davidson and a Williamson matter. Middle Tennessee's ring — Davidson, Williamson, Rutherford, Wilson, Sumner, Robertson, Cheatham, Maury — is 8 counties, ~1.3M rooftops, and the densest seams in the state. Sign those and the map sells itself to the next ring.

Then the other three metros (Knox, Hamilton, Shelby) — each is one county plus a city with its own open data — then the rest by seam density.

What “sign up” looks like

1. A finance director opens their county's free boundary report. It shows the seams, the exposure ranking, the State-data conflicts, and the count of named businesses in each class.
2. They click “Start monitoring.” Fixed annual fee under the bid threshold (pricing memo). An engagement letter with the data-handling protocol and the client-furnishes-GIS clause.
3. They request their situs report through REP (we provide the request; they send it).
4. First roster join, adjudicated queue, correction packets — within two weeks of the roster.
5. Quarterly thereafter: refresh, re-measure, new/resolved/moved, packets, bulletin.
6. Annually: the license and TPP check against the same placed universe.

What could stop it

- **§ 67-1-1704(e) reuse across jurisdictions** — get counsel's answer before the second county signs. The public surface is unaffected; the question is only about roster-derived facts.
- **Financial Control's three unanswered questions** — one phone call.
- **Business universe quality** — the SOS file has holding companies and stale addresses; the calibration loop is what keeps this honest, and it should be published to every client.
- **Adjudication discipline** — the felony exposure is personal. Two or three named people, a written protocol, no shortcuts.